

Team 3 Music App Final Report

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I. INTRODUCTION

Our group was tasked with creating a unique application user interface from the ground up, with the goal of the project being to learn about how to create and iterate through the early stages of GUI development.

The app that has been developed is a conceptual music app, with dummy function merely for the purpose of user testing. The app is meant to be a place for universal music storage. It allows the user to search for and find songs from any streaming service that they have signed into and store them all in the same place. Meaning that they will not have to switch apps if they want to listen to something specific.

The main target audience for this app is dedicated music listeners who are into more "underground" music, since many indie artists may not be on multiple streaming platforms.

II. SYSTEM DESCRIPTION

The app is designed for mobile devices and follows the common layout that most modern music apps do. It has a main page for the currently playing music and the queue, a search page for the user to find new music, and a library page for the user to manage and organize their selections.

Where our app begins to differ from other apps is in the more simple design of the user interface. We tried to create the most minimal user interface possible while keeping things usable. Since our app only makes use of songs or albums selected by the user, there is no space wasted with pages such as radio, podcasts, or hot new music.

A. Music Bar

The Music Bar is always visible when using the app, and houses critical buttons for navigation between main pages, as well as music information and controls. It displays information like the song title and the artist, as well as having the skip, rewind, and pause buttons to the right.

B. Now Playing Page

The Now Playing page of our app is what first greets the user when they open the application. Sitting above the Music Bar is a visual element that displays the cover art for whatever is playing, while also acting as the button to access the queue.

We had originally wished to add a few more visually pleasing elements that related to the queue, but ended up not having the time during our development stages. Instead, we focused on other more critical areas of the app.

C. Queue Page

The Queue page holds a list of all the songs that the user has queued. The song at the bottom is highlighted orange to show the user that it is what is playing at the moment. While the songs above are listed in order of what is to play next. The queue can be rearranged by use of buttons to the right of each song.

D. Search Page

The Search page is what allows the user to find their favorite music and add it to their collection. At the top of the page there is a search bar where the user can enter the name of any song they would like. Once something has been entered into the search bar, the results will populate below it.

Each song that appears will display its name and a button to add the song on the right. To the left of name the art for the song will be shown. The art also functions as a button to "quick queue" the song. This button will add the song to the users collection, as well as the queue.

E. Collection Page

The Collection page is where the user can access and organize all of their selected music. The basic view of the Collection page shows the user a 2x1 scroll view of all of their music. The music on the page will be outlined with a color of the original streaming service. At the top of the page is the button to create a new playlist; this will be discussed later.

When selecting music on the Collection page, the user will be brought to another screen containing more details such as the artist, song or songs, and some buttons to either play or queue the music. The user can queue the whole collection or just individual songs.

F. Playlist Creation Page

The Playlist Creation screen starts off with a view similar to that of an already saved collection, but with no data present. The user can type a name into the text box, then click the "+" button to add songs to it. To add songs, the user can scroll through their saved songs and press the button on the side to add it to the playlist. When they user is done they can click "Done" at the top and they are brought back to the creation page, where they can see the same and songs that have been added. To finalize the playlist there is a check button to save the playlist to the users collection.

III. METHODS

A. Participants

For our user study we had a total of 10 participants, ages ranging from 18 to 56. Participants were mostly friends and family, with a few of them being university students. Some of the participants were used to frequently using music apps, and others rarely did. Overall, the range of participants in our user study represents a large user base and is not limited to a single demographic.

B. Experimental Design

Our user testing was carried out as a within subjects study. Each participant was assigned the same tasks and the same questions to fill out. We did this to minimize the number of participants needed, and to have a single testing scenario instead of needing multiple surveys.

C. Hypotheses

Our first hypothesis for our user study was that "users can perform common tasks in the same or a smaller number of clicks as competing apps".

The second hypothesis was that "users will find the apps user interface easier to interact with than that of competing apps".

The data that we chose to collect for the study was selected so that we could analyze it to see if our hypothesis were true or not. We also chose to have the participant complete two rounds of trials, one using our own app, and another using an music app of their choice.

D. Apparatus

Setup for the user trials was straightforward. The trial admin would run the app on their machine, and open the survey in another tab. The participant is briefed with basic data of the apps purpose before the test begins.

The participant will work through the tasks one by one, while the admin fills out a form for the tasks. Each task is timed and the number of actions taken by the participant are counted. Once the task is complete the admin asks the user some questions about the task and records the responses. There are a total of six tasks for the participant to complete. After all of the tasks have been completed the user will answer questions about the app using a Likert Scale.

The list of tasks guides the user through the app and allows them to experience and use all of the features at least once. Tasks start out with smaller single goals, then grow to larger proportions later in the list after the user got more acclimated to the app.

IV. RESULTS

The main statistics that we looked at from our survey data were the time to complete each task, the number of actions per task, how easy the apps features were to understand, and how easy the app was to navigate.

Our testing showed that it almost always took participants longer to complete tasks, and took around the same number

of actions to complete tasks as the competing apps. The data in Figs. 1 and 2 below represent the average time or actions for each task and are grouped per app that was tested.



Fig. 1. Average task completion time between apps

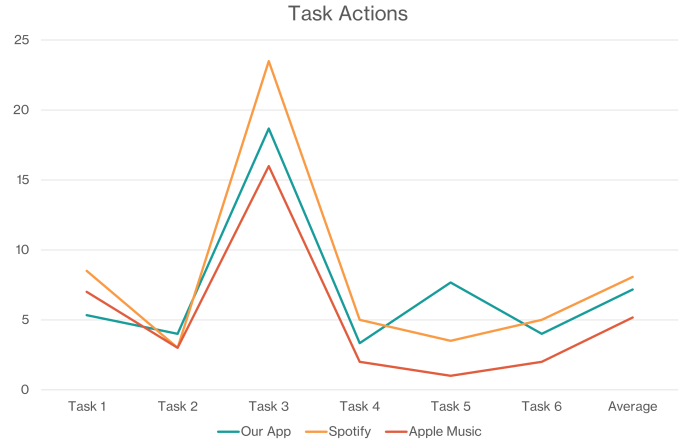


Fig. 2. Average number of actions per task between apps

Regarding the mean number of actions for the tasks, our app had a slightly lower number of actions than the competing apps. Our app had a mean of 7.22 actions for all tasks, and the competing apps had a mean of 7.44. The p value of the means is roughly 0.93, which means that there is not a significant difference in the mean number of actions for tasks throughout all three apps tested.

In terms of our qualitative results, participants often felt less comfortable navigating our app and using its features. Figs. 3 and 4 show the percentages of different responses from the Likert scale section of our survey.

V. DISCUSSION

The results from our user trials were different from what we were initially expecting. Upon first glance, it may seem like our app severely underperformed its competitors in terms of task completion time. But we have concluded that the large

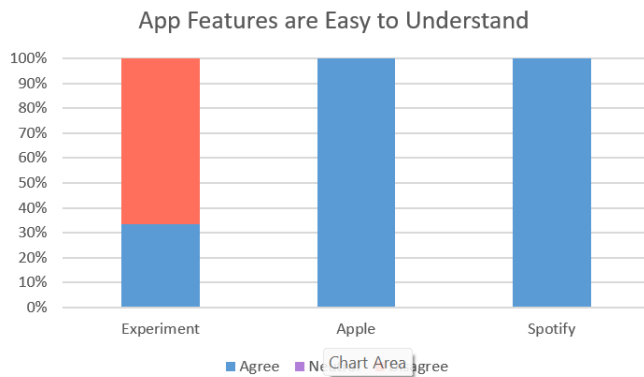


Fig. 3. "The apps features are easy to understand" responses

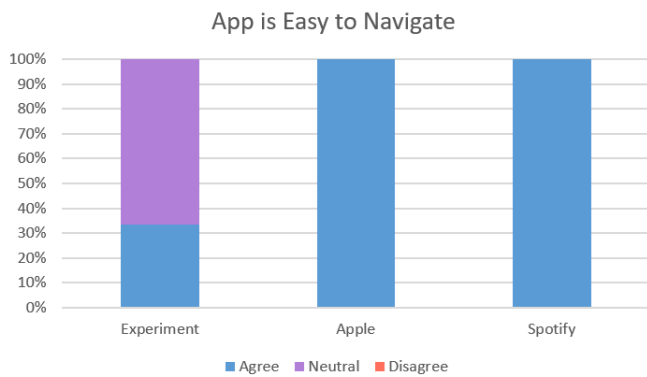


Fig. 4. "The app is easy to navigate" responses

difference is due to the other app the participants tested with being one they were already familiar with.

Our app performed within the averages of the other apps tested when it came to the number of actions per task. This is good as it shows that with experience users should be able to use our app in a similar time frame to competing apps, since the mean number of actions taken for tasks is almost the same as competing apps.

Participants felt less confident when using our app compared to others, based on responses from the Likert scale section of our survey. We think this is again due in part to the user being familiar with the competing app but not our own app.

After looking at the data shown in Section IV, we have concluded that there are multiple factors that have contributed to some errors in our data. If we are to go through with more trials, we will make sure that participants test with apps that they are not already proficient with. We have also considered testing users using the System Usability Scale (SUS), where we would be able to collect data on not only the usability of our app but also the learnability, as stated by Lewis and Sauro [1]. This should help to balance the gap in familiarity and yield more accurate results, as well as allowing us to see if our app can be learned by users faster than competing apps.

VI. CONCLUSIONS

Overall, the development process produced a usable interface in the end. Each new phase of user testing allowed us to improve our designs and create a product that users were able to interact with and use with only minor issues.

Apart from the paper prototype testing phase, our app could have benefited from better testing methods. There were a few factors in our testing procedures that led to results that make our app appear to be under performing compared to competitors. However, our app did meet at least one of our expectations by matching the mean number of actions for tasks. If given another chance to iterate testing, we would proceed with better testing methods and be able to refine our app further.

VII. REFERENCES

REFERENCES

- [1] J. R. Lewis and J. Sauro, "The factor structure of the system usability scale," in *Proceedings of the 1st International Conference on Human-Computer Interaction*, Berlin, Heidelberg: Springer, 2009, p. 9.